

# Ahmad Mustafa Anis

+92-314-5617800 | [ahmadmustafaanis.github.io](https://ahmadmustafaanis.github.io) | LinkedIn: [// ahmad-anis](https://www.linkedin.com/in/ahmad-anis) | GitHub: [// ahmadmustafaanis](https://github.com/ahmadmustafaanis) | Email: [// ahmadanis5050@gmail.com](mailto:ahmadanis5050@gmail.com)

## EDUCATION

**Carnegie Mellon University**  
*Masters of Science in Artificial Intelligence Engineering*  
**International Islamic University**  
*Bachelors of Science in Computer Science*

Pittsburgh, PA  
Aug 2026 – Dec 2027  
Islamabad, Pakistan  
September 2018 – September 2022

## EXPERIENCE

**Research Assistant** June 2026 – Current  
*TarrLab — CMU*

- Working on Self Supervised Learning, supervised by [Professor Michael Tarr](#) and [Professor Deva Ramanan](#).

**Deep Learning Engineer** Mar 2024 – Aug 2026  
*Roll AI*

- Optimized real-time video bokeh effects by implementing Depth Estimation and Matting algorithms, achieving 2x performance improvement through parallel processing and batch optimization for live streaming applications
- Developed and Deployed Active Speaker Detection system for multi-camera live events, automating speaker-camera associations and reducing event video editing time from 48 hours to 1 hour (98% improvement)
- Developed a Video Understanding pipeline for automated chapter generation and captioning, enabling intelligent b-roll suggestions by analyzing semantic content and scene transitions across long-form video.
- Developed a multi-view video generation pipeline that synthesizes novel viewpoints from temporal endpoints, enabling enhanced scene reconstruction for live event production.
- Architected a multimodal LLM Agent as a centralized AI video editing system, automating social media clip generation, multi-take video orientation/cropping, transcript correction and enhancement, and dynamic VFX/motion graphics creation to streamline post-production workflow.

**AI Research Fellow** Sep 2024 – June 2025  
*Fatima Fellowship*

- Working on Retrieval Augmented Test Time Inference for Medical Vision Language Models, advised by [Dr. Wei Peng](#) from Stanford University

**Machine Learning Engineer** April 2022 – Mar 2024  
*Red Buffer*

- Created a fact verification and identification pipeline, integrated multi-threading for a 200% speed increase, fine-tuned LLaMAv2, Dolly-3B, GPT-Neo, and BART for negation generation to search for false news.
- Improved OCR latency from 0.6 FPS to 16 FPS on the CPU for streaming devices to blur sensitive information.
- Boosted Accuracy to 85% from 30% for logo recognition during cricket matches using CLIP by OpenAI.
- Developed a drag and drop RAG system which can be used with any LLM, vector DB, and multi-hop search.

**AI Fellow** December 2023 – February 2024  
*PI School of AI*

- I was selected (10 in 300 applicants) from all over the world to turn with fellowship worth 12,500€ where I worked on extracting key information from long context interview transcripts using LLMs (**MistralLite**, **Mixtral8x7B**). Worked on RAG using **DSPy**.

**Deep Learning and Computer Vision Intern** July 2021 – Dec 2021  
*WortelAI*

- Developed Safety Clothes Detection for Factory workers using YOLOv5.
- Trained a Celebrity Detection and Classification system using 3DDFAv2, ResNet50, and a 172GB Dataset.
- Created a real-time search engine for videos using CLIP by OpenAI, Qdrant for Vector DB, and ElasticSearch.
- Performed Multi-label Text Classification using BERT and PyTorch for 1500+ labels.

## OPEN SOURCE CONTRIBUTIONS

[Pandas](#), [LLama-Index](#) [Google Research BigVision](#), [OpenAI Shap-E](#),

## TECHNICAL SKILLS

---

**Programming Languages:** Python, C++, Rust, Java, JavaScript

**Libraries/Frameworks:** PyTorch, Tensorflow, PyTorch Lightning, Distributed Training, OpenCV, Flask, FastAPI

**Databases:** MySQL, PostgreSQL

**Tools:** Git, GitHub, GitLab, Docker, Markdown, Jupyter Notebook, Anaconda, Visual Studio Code, Pep8

## PUBLICATIONS

---

- **A. M. Anis**, Hasnain Ali, M. Saquib Sarfraz *On the Limitations of Vision-Language Models in Understanding Image Transforms*, **CVPR 2025w: Computer Vision in the Wild** [Paper].
- S. Longpre, N. Singh,... **A. M. Anis**(11th Author), et al. *Bridging the Data Provenance Gap Across Text, Speech, and Video*. **ICLR 2025**.
- S. Longpre, R. Mahari,... **A. M. Anis**(16th Author), et al. *Consent in Crisis: The Rapid Decline of the AI Data Commons*. Presented at **NeurIPS 2024**. [Paper]

## SELECTED PROJECT

---

### DistAYA

- [AYA Expedition](#) project, creating small multilingual LLMs distilled from large multilingual LLMs, [Hugging Face](#)

## VOLUNTEER EXPERIENCE

---

**Research Contributor** | [Data Providence Initiative](#) | *MIT Media Lab* Dec 2023 – Present

- Auditing LLM training data to assess web data access shifts, cultural representation and advocating for transparency.

**Volunteer Researcher & Regional Community Lead** | [Cohere Labs Community](#) Nov 2022 – Present

- [Organized over 50 guest speaker sessions with Asian researchers from top universities and institutes.](#)
- Initiated NYU Deep Learning study group (Prof. Yann LeCun).
- Urdu Language lead for AYA, a Multi-Lingual model by Cohere. Contributed 3 Urdu datasets for AYA collection.

**Mentor** | [MESA program at Foothill College](#) | *MESA@Foothill* Jan 2024 – Present

- Providing mentorship to underrepresented students in STEM at Foothill College.

**Fellowship Lead** | [Bytewise Limited](#)

- NLP Lead (Jun-Aug 2024): Mentoring 20 students, supervising the NLP curriculum.

**AWS Community Builder** | *Amazon Web Services* Aug 2022 – Apr 2023

- Organized events to raise awareness about AWS services in local community.

## INTERVIEWS , PODCASTS, BLOGS & TECHNICAL WRITING

---

[Interview with AtomCamp](#) providing advice for beginners in Machine Learning.

[Podcast with Adventures in Machine Learning](#) on how to optimize and improve **OCR** systems

[Podcast with Adventures in Machine Learning](#) on how to deploy Deep Learning models with **FastAPI**

[Medium](#), [KDNuggets](#), [CNVRG](#)

## CERTIFICATIONS & ACHIEVEMENTS

---

### Courses & Certifications:

- Microsoft Technology Associate (MTA) 98-381 Introduction to Programming Using Python
- Machine Learning by Coursera
- Deep Learning Specialization by Coursera
- Getting Started with TensorFlow 2 by Coursera
- NYU Deep Learning by Yann LeCun
- Mathematics for Machine Learning by Imperial College London

### Achievements:

- [2600+ points on StackOverflow](#), impacting 300k+ people
- [5 kyu, top 36% worldwide at Codewars](#)
- [Platinum badge on KDNuggets](#)